Chapter test

29 Biodiversity

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Which of the following shows the hierarchy of taxa in order of increasing diversity?

   A. genus, family, order, class
   B. family, genus, class, order
   C. genus, family, class, order
   D. class, order, family, genus ☐

2. Based on binomial nomenclature, which two species are most closely related?

   (1) Hong Kong camellia (*Camellia hongkongensis*)
   (2) Hong Kong balsam (*Impatiens hongkongensis*)
   (3) Hong Kong pencilwood (*Dysoxylum hongkongense*)
   (4) Hong Kong azalea (*Rhododendron hongkongense*)
   (5) Taiwan azalea (*Rhododendron oldhamii*)
   (6) giant timber bamboo (*Bambusa oldhamii*)

   A. (1) and (2)
   B. (3) and (4)
   C. (4) and (5)
   D. (5) and (6) ☐

★ 3. To claim that a group of organisms represents a new species, biologists must show that the organisms

   A. look similar to one another.
   B. look different from known species.
   C. have similar genetic composition to known species.
   D. do not produce fertile offspring with known species. ☐

© 2020 Aristo Educational Press Ltd.

4. Which of the following statements is/are correct?

(1) Archaebacteria are more closely related to eukaryotes than to bacteria.
(2) All prokaryotes are bacteria.
(3) Fungi are more closely related to plants than to animals.

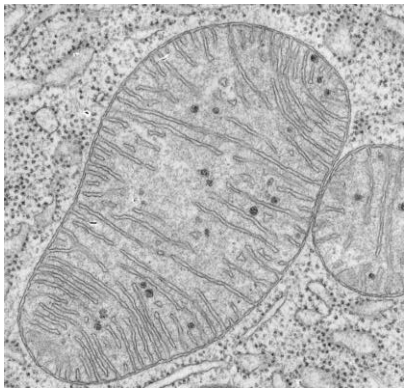
A. (1) only
B. (2) only
C. (1) and (3) only
D. (2) and (3) only ☐

5. Which of the following structure(s) is/are present in both prokaryotic cells and plant cells?

(1) a cell wall
(2) ribosomes
(3) chloroplasts

A. (1) only
B. (1) and (2) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

★ 6. The electron micrograph below shows part of a cell.



The cell is probably taken from

(1) a protist.
(2) a bacterium.
(3) a plant.

A. (1) only
B. (2) only
C. (1) and (3) only
D. (1), (2) and (3) ☐

© 2020 Aristo Educational Press Ltd.

7. The photograph below shows some organisms living on a tree log.



Which of the following is ***not*** a characteristic of these organisms?

A. They make their own food.
B. They reproduce by spores.
C. Their cells have cell walls.
D. They have no roots. ☐

★ 8. The photographs below show three different plants, X, Y and Z.



X



Y



Z

What do plants X, Y and Z produce for reproduction?

	X	***Y***	***Z***
A.	spores	spores	seeds
B.	spores	seeds	spores
C.	seeds	spores	seeds
D.	seeds	seeds	spores ☐

© 2020 Aristo Educational Press Ltd.

Directions: Questions 9 and 10 refer to the table below, which shows the features of four marine animals, P, Q, R and S.

Feature	P	Q	R	S
Has vertebral column	✓	✓	×	✓
Has gills	✓	×	✓	×
Has lungs	×	✓	×	✓
Body covered with scales	✓	✓	×	×
Has mammary glands	×	×	×	✓

Key: ✓ = present × = absent

9. Which of these organisms could be a turtle?

A. P
B. Q
C. R
D. S ☐

★ 10. S is classified as a mammal but not a fish because

A. it does not have gills.
B. it has lungs.
C. it's body is not covered with scales.
D. it has mammary glands. ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	29.2	29.2	29.2	29.2	29.3	29.3	29.3	29.3	29.3	29.3

© 2020 Aristo Educational Press Ltd.

B. Problem solving (23 marks)

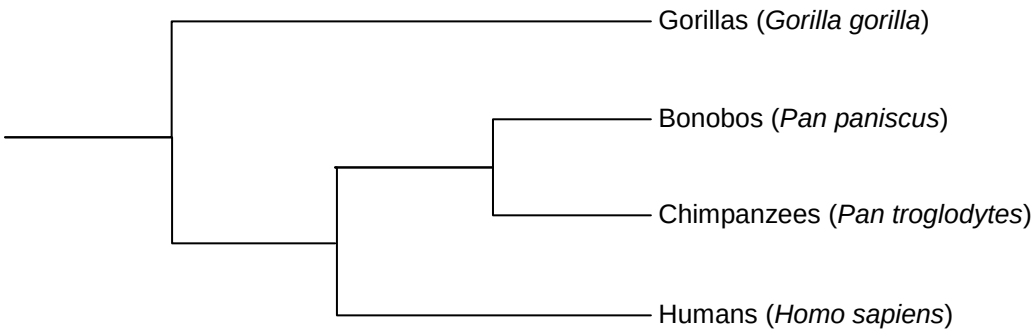
11. The table below show the taxonomic groups to which humans (*Homo sapiens*) belong.

Taxonomic group	Name	Sequence
Kingdom	Animalia	
Genus		
Species		
Phylum	Chordata	
Class	Mammalia	
Order	Primates	
Family	Hominidae	

(a) Complete the first two columns of the table. (4 marks)

(b) Write the numbers 1 to 8 in the third column to show the sequence of taxonomic groups. Write the number 1 to show the group with the most organisms and the number 8 the least. (1 mark)

The diagram below shows the classification of some organisms in the family Hominidae.



(c) Gorillas, bonobos, chimpanzees and humans all belong to the same family, Hominidae. Name another taxonomic group to which all these organisms belong. (1 mark)

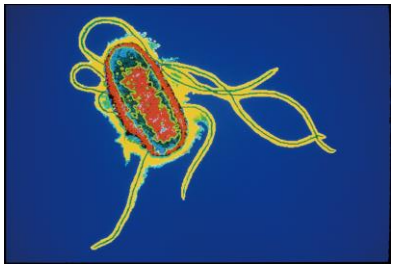
_____

© 2020 Aristo Educational Press Ltd.

(d) Using evidence from the diagram, state and explain which two organisms are most closely related. (2 marks)

(Total: 8 marks)

★ 12. The photographs below show three different organisms.



X



Y



Z

(a) State ***two*** cell structures that can be found in Y and Z, but not in X. (2 marks)

(b) Organisms Y and Z belong to the kingdom Protista. Based on the photographs, suggest one reason why Y and Z are classified in the same kingdom. (1 mark)

(c) In the past before the kingdom Protista was created, organisms were classified into two kingdoms, either Plantae or Animalia.

Based on the photograph, suggest why Z could be classified into

(i) Plantae. (1 mark)

(ii) Animalia. (1 mark)

© 2020 Aristo Educational Press Ltd.

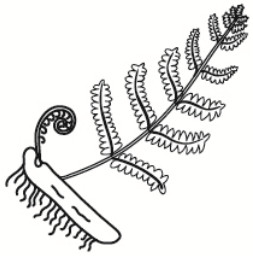
(d) Viruses are not classified into a kingdom because many scientists do not consider them as organisms. Give ***one*** reason for and against the view that viruses are organisms. (2 marks)

(Total: 7 marks)

13. The diagrams below show six different organisms (not drawn to the same scale).



A



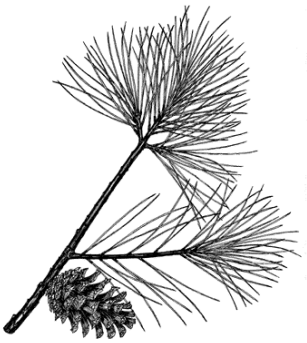
B



C



E



F



G

© 2020 Aristo Educational Press Ltd.

(a) Organisms B, C, D, E, F and G belong to the kingdom Plantae. Give ***two*** common features of members of this kingdom. (2 marks)

(b) Give ***two*** features that distinguish E from other members of the kingdom Plantae. (2 marks)

(c) C and G belong to the same sub-group while F belongs to another sub-group. Give ***one*** feature that distinguishes these two sub-groups. (1 mark)

(d) Organism A was once classified in the kingdom Plantae and was later reclassified in a different kingdom.

(i) According to six-kingdom system, name the kingdom to which A belongs. (1 mark)

(ii) Suggest a reason why A was classified in the kingdom Plantae. (1 mark)

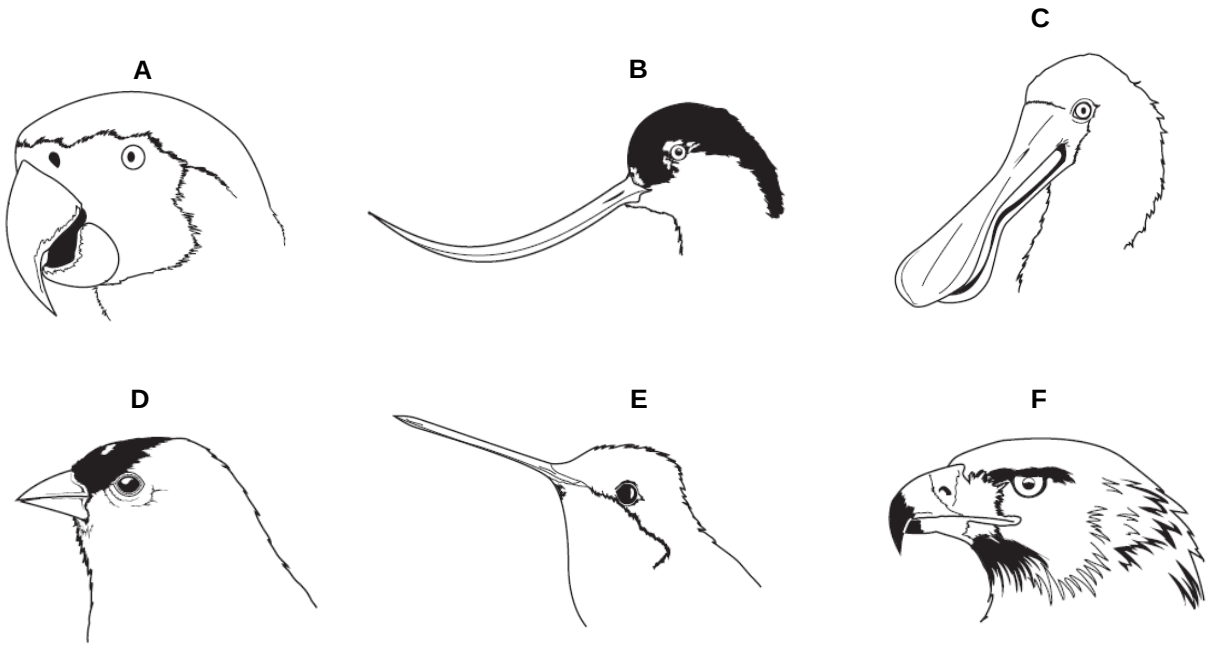
(iii) Suggest a reason why A was separated from the kingdom Plantae and reclassified. (1 mark)

(Total: 8 marks)

© 2020 Aristo Educational Press Ltd.

C. Exam challenge (7 marks)

★★ 14. The diagrams below show the heads of six different birds (not drawn to the same scale).



(a) In the table below, put a '✓' in the appropriate box to show the relative size and shape of the beak of each bird. (2 marks)

	Beak shorter than head	Beak longer than head	Beak straight	Beak curved	Other features
A					Upper mandible longer than lower mandible; lower mandible about half the length of the upper mandible
B					Beak curves upwards
C					Both mandibles widen at the end of the beak
D					Upper and lower mandibles same length
E					Both mandibles similar width along the whole length
F					Upper mandible longer than lower mandible; lower mandible more than half the length of the upper mandible

© 2020 Aristo Educational Press Ltd.

(b) Using the information from the table in (a), complete the following dichotomous key.

(3 marks)

1a	Beak is shorter than the head		go to **2**
1b	Beak is longer than the head		go to **4**
2a		.....	**D**
2b		.....	go to **3**
3a	Lower mandible is about half the length of the upper mandible		
3b	Lower mandible is more than half the length of the upper mandible		
4a	Both mandibles are widen at the end of the beak		**C**
4b	Both mandibles have a similar width along the whole length		go to **5**
5a		.....	**E**
5b		.....	**B**

(c) Identify, with a reason, which of the birds A to F is adapted to feed on nectar from flowers. (1 mark)

_____

_____

(d) Apart from the beak, give another distinguishing feature of birds. (1 mark)

_____

(Total: 7 marks)

© 2020 Aristo Educational Press Ltd.

Marking scheme

A. Concept checking (10 marks, 1 mark each)

1.	A	2.	C	3.	D	4.	A	5.	B
6.	C	7.	A	8.	C	9.	B	10.	D

(1 mark each)

B. Problem solving (23 marks)

11. (a), (b)

Taxonomic group	Name	Sequence
Kingdom	Animalia	2
Genus	*Homo*	7
Species	*Homo sapiens*	8
Phylum	Chordata	3
Class	Mammalia	4
Order	Primates	5
Family	Hominidae	6
Domain	Eukarya	1

(5)

(c) Domain / kingdom / phylum / class / order (1)

(d) Bonobos *(Pan paniscus)* and chimpanzees (*Pan troglodytes*) (1)
They belong to the same genus while other organisms belong to different genera. /
They have a more recent common ancestor than any other two organisms shown. (1)

(Total: 8 marks)

12. (a) Nucleus / mitochondria / endoplasmic reticulum (any ***two***) (1, 1)

(b) Both are unicellular and eukaryotic / have a nucleus. (1)

(c) (i) It contains chlorophyll for photosynthesis. (1)

(ii) It has a flagellum for swimming / locomotion. (1)

(d) For: Viruses can reproduce in a cell / have genetic materials to control life processes. (1)
Against: Viruses are non-cellular / have no cytoplasm / cannot replicate by themselves / have no metabolism unless they are within a host cell. (1)

(Total: 7 marks)

© 2020 Aristo Educational Press Ltd.

13. (a) Any ***two*** of the following: (1, 1)
    - They are multicellular. (1)
    - They are eukaryotic. (1)
    - They are autotrophs / have chlorophyll for photosynthesis. (1)
    - Their cells have a cell wall composed of cellulose. (1)

    (b) Any ***two*** of the following: (1, 1)
    - It has no vascular tissues. (1)
    - It has no true roots. (1)
    - It has rhizoids for anchorage and for absorption of water and minerals. (1)

    (c) Plants in the sub-group of C and G produce flowers for reproduction while those in the sub-group of F do not. / Plants in the sub-group of F bear cones which carry seeds for reproduction. (1)

    (d) (i) Fungi (1)

    (ii) Both A and plants do not move about / have cell walls. (1)

    (iii) A does not carry out photosynthesis but plants do. / The composition of the cell wall of A is different from that of plants. (1)

(Total: 8 marks)

© 2020 Aristo Educational Press Ltd.

C. Exam challenge (7 marks)

14. (a)

	Beak shorter than head	Beak longer than head	Beak straight	Beak curved	Other features
A	✓			✓	Upper mandible longer than lower mandible; lower mandible about half the length of the upper mandible
B		✓		✓	Beak curves upwards
C		✓	✓		Both mandibles widen at the end of the beak
D	✓		✓		Upper and lower mandibles same length
E		✓	✓		Both mandibles similar width along the whole length
F	✓			✓	Upper mandible longer than lower mandible; lower mandible more than half the length of the upper mandible

(1) (1) (2)

(b) **2a** Upper and lower mandibles are the same length
2b Upper mandible is longer than the lower mandible (1)
3a A
3b F (1)
5a Beak is straight
5b Back is curved / curves upwards (1)

(c) E. The beak is long and narrow that can probe deep into flowers to extract nectar. (1)

(d) Feathers (1)

(Total: 7 marks)

© 2020 Aristo Educational Press Ltd.